# 沐曦 XTuner 微调 InternLM 赛事 Baseline

## 1. 操作视频

https://aicarrier.feishu.cn/minutes/obcni337zj434hm7175kz3k8?from=from_copylink

## 2. 环境介绍

GPU：沐曦MXC500（32G）

CPU：8核心

内存：96G

## 3. DaoCloud平台操作

上海六区地址：https://console.d.run/zestu/market?regionId=sh-06

### 3.1 初始化存储

操作步骤：文件存储—>选择《上海六区》->初始化文件存储

说明：文件存储将会挂载在/root/data 目录下，免费容量为20G

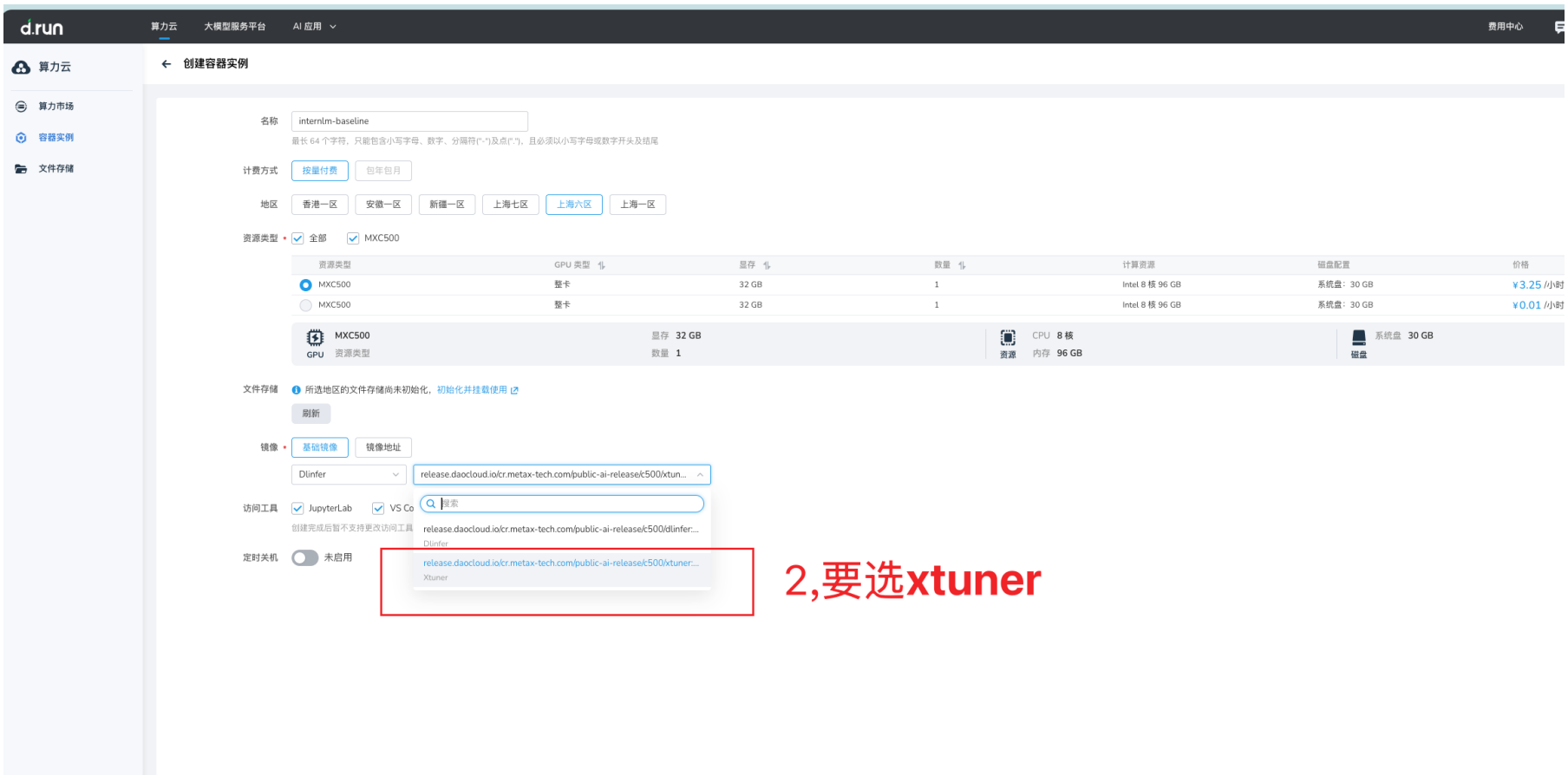
特别注意：初始化完成后，请在web页面上传一个空文件,否则无法挂载成功

### 3.2 选择算力云->算力市场->上海六区（如下图选择MXC500）立即购买

### 3.3 选择镜像

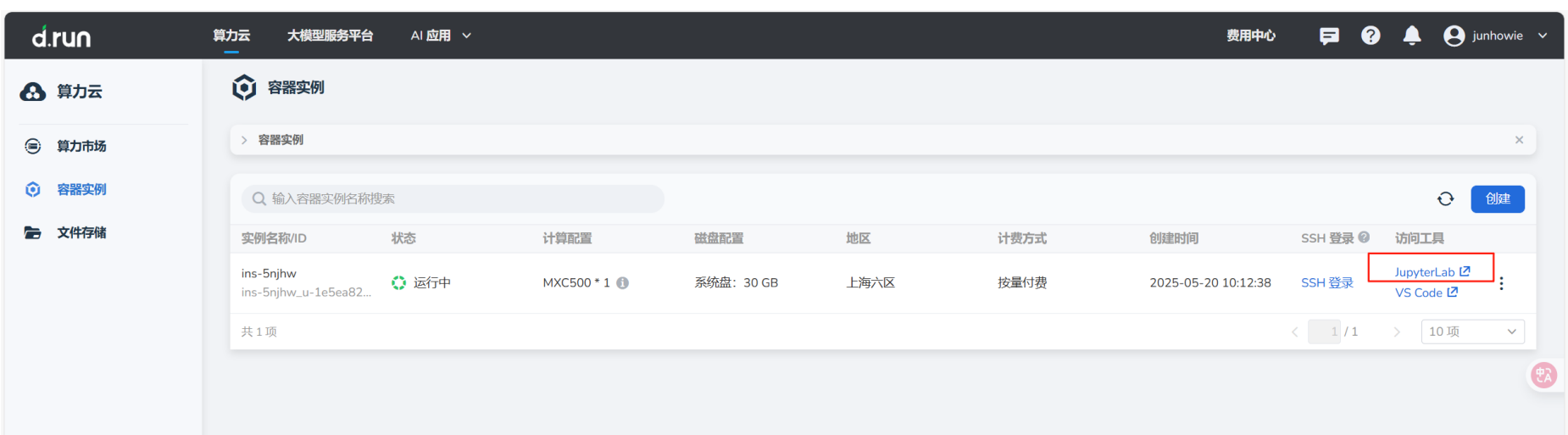
下拉框1：基础镜像->Dlinfer

下拉框2：选择带有Xtuner的镜像版本（如下图）




## 3.4 创建并启动容器实例

1.选择JuypterLab 进入开发机



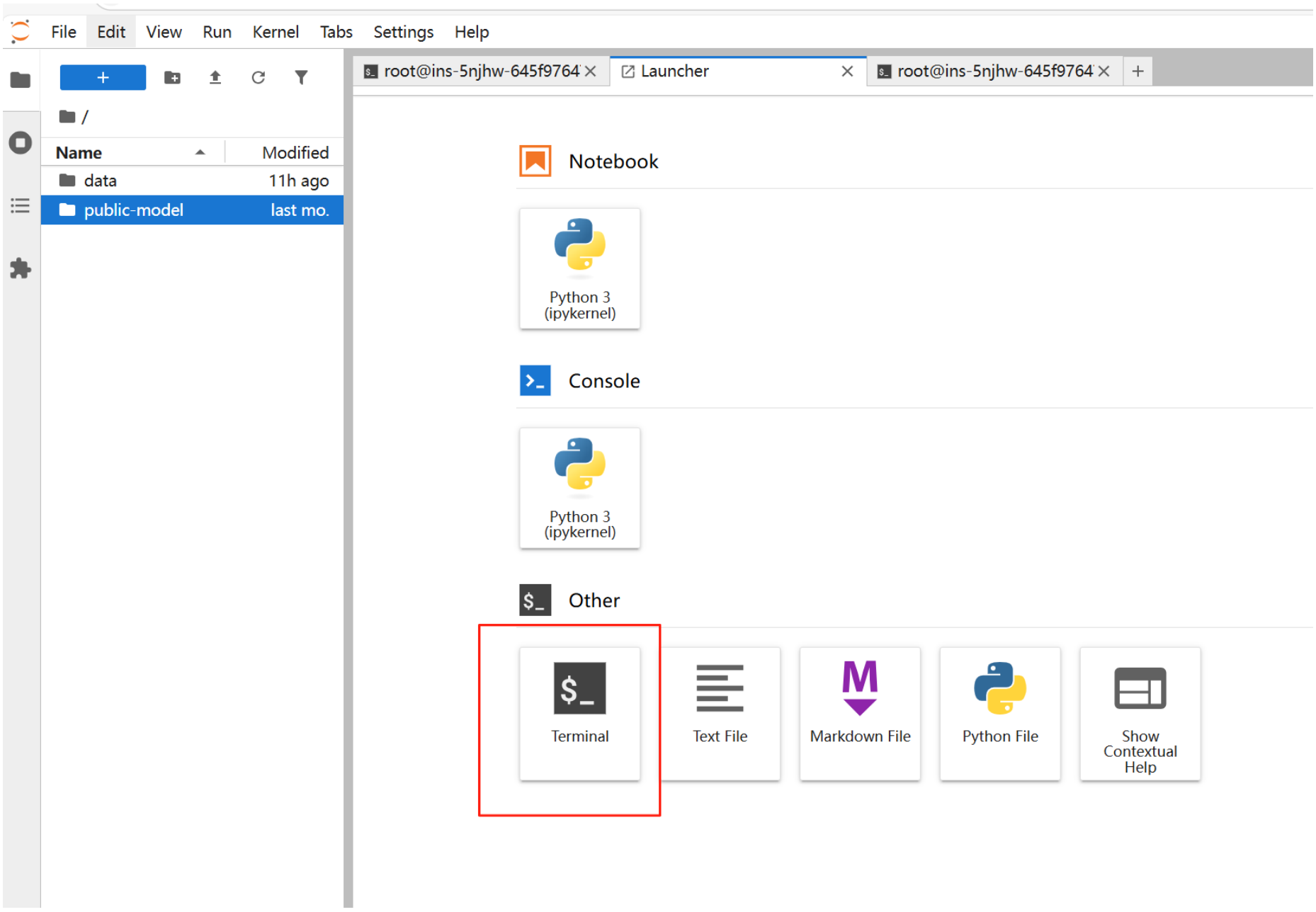

# 4. 开发机介绍

## 4.1 文件夹介绍

默认进入/root 目录下

data文件夹为初始化挂载文件夹（读写）

Public-model 为平台共享模型仓库




## 4.2 终端介绍

```
温馨提示:
+----------------------+----------+-------------------------------------------------------------------------------------+
        目录          |   名称   |                                                 说明
+----------------------+----------+-------------------------------------------------------------------------------------+
  /                    |  系统盘  | 安装及配置环境的目录，可存放代码等。会随着释放实例消失，当前暂不支持保存系统盘
  /root/data           |  文件存储 | 用于存储数据的目录，支持持久化，需确保正确挂载
  /root/public-model   |  模型文件 | 平台在容器实例中提供公开的模型文件，用户可直接调用，快速部署 AI 应用
+----------------------+----------+-------------------------------------------------------------------------------------+

  -------- 系统信息 --------

  OS 版本  : Ubuntu 22.04.3 LTS
  内核      : 5.15.0-58-generic
  IP 地址     : 10.233.112.76
  主机名    : ins-5njhw-645f976478-djpwb

  CPU 型号   : Intel(R) Xeon(R) Platinum 8460Y+
  CPU 线程 : 8 C
  内存      : 139 MB / 98304 MB (0.14% 已使用)
  GPU         : NO GPU detected
  CUDA        : NO CUDA detected

  ------ 文件系统信息 ------

  挂载路径: /root/data    6.7G / 20G (34% 已使用)

root@ins-5njhw-645f976478-djpwb:~#
```

## 4.3 环境检查

镜像中已经内置Xtuner

pip show

```
pip show xtuner
```

```
#可以看到环境中已经有了xtuner
#Name: xtuner
#Version: 0.1.23
#Summary: An efficient, flexible and full-featured toolkit for fine-tuning large models
#Home-page: https://github.com/InternLM/xtuner
#Author: XTuner Contributors
#Author-email: openmmlab@gmail.com
#License: Apache License 2.0
#Location: /workspace/xtuner
#Editable project location: /workspace/xtuner
#Requires: bitsandbytes, datasets, einops, lagent, mmengine, openpyxl, peft, scikit-image, scipy, SentencePiece, tiktoken, torch, torchvision, transformers, transformers_stream_generator
#Required-by:

```

## 4.4 GPU信息监控

mx-smi 与 nvidia-smi 的功能一致

mx-smi

```
root@ins-5njhw-645f976478-djpwb:~# mx-smi
mx-smi  version: 2.1.9

=================== MetaX System Management Interface Log ===================
Timestamp                                         : Tue May 20 10:48:51 2025

Attached GPUs                                     : 1
+-----------------------------------------------------------------------------+
| MX-SMI 2.1.9                        Kernel Mode Driver Version: 2.9.8       |
| MACA Version: 2.31.0.6              BIOS Version: 1.20.3.0                  |
|-------------------------------------+----------------------+----------------+
| GPU         NAME                    | Bus-id               | GPU-Util       |
| Temp        Power                   | Memory-Usage         |                |
|=====================================+======================+================|
| 0           MXC500 VF               | 0000:3b:00.2         | 0%             |
| N/A         N/A                     | 618/32768 MiB        |                |
+-------------------------------------+----------------------+----------------+

+-----------------------------------------------------------------------------+
| Process:                                                                    |
|  GPU                   PID         Process Name                 GPU Memory  |
|                                                                 Usage(MiB)  |
|=============================================================================|
|  no process found                                                           |
+-----------------------------------------------------------------------------+

End of Log
root@ins-5njhw-645f976478-djpwb:~#
```

# 5. 补充安装包

## 5.1 安装mpi4py

Code block

```
pip install mpi4py

pip install transformers==4.48.0
```

建议安装，否则报错

## 5.2 链接模型位置

Code block

```
ln -s /root/public-model/models/Shanghai_AI_Laboratory/internlm2_5-1_8b-chat ./
```

## 5.3 数据集（已经处理好）

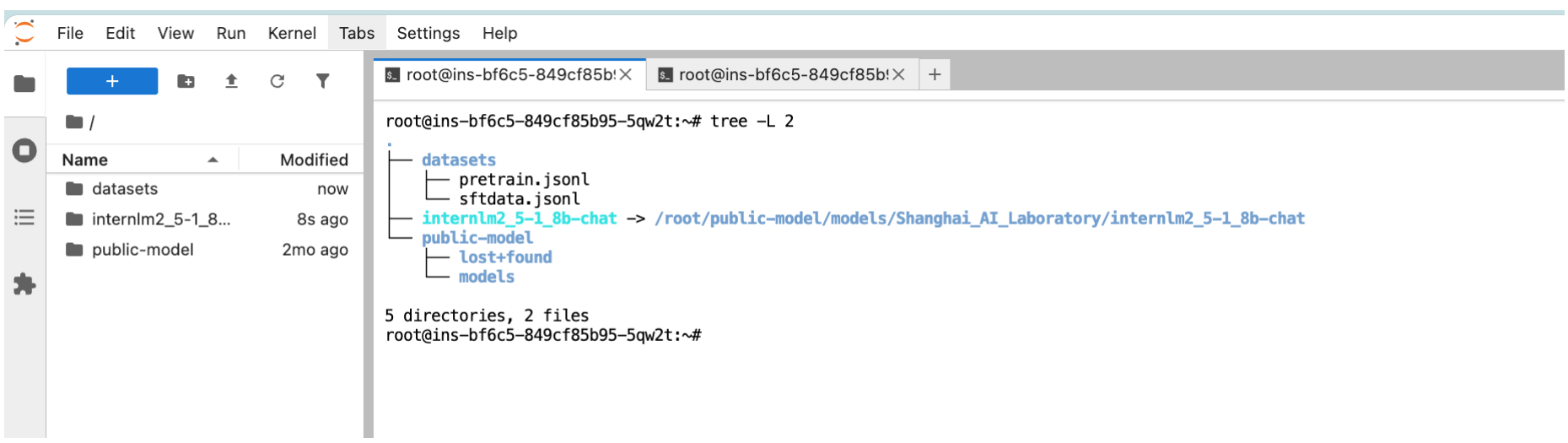
pretrain.jsonl
798.03KB

sftdata.jsonl
3.00MB

Code block

```
apt install tree
tree -L 2
```

可以将项目目录结构组织成如下图所示：




# 6. 开启训练

## 6.1 预训练

Code block

```
xtuner train pretrain.py --deepspeed deepspeed_zero1
```

注意替换模型和数据集的位置分别位于23行27行

Code block

```
1  pretrained_model_name_or_path = './internlm2_5-1_8b-chat'
2
3  data_files = ['./datasets/pretrain.jsonl']
```

pretrain.py

```
1  import torch
2  from datasets import load_dataset
3  from mmengine.dataset import DefaultSampler
4  from mmengine.hooks import (CheckpointHook, DistSamplerSeedHook, IterTimerHook,
5                              LoggerHook, ParamSchedulerHook)
6  from mmengine.optim import AmpOptimWrapper, CosineAnnealingLR, LinearLR
7  from peft import LoraConfig
8  from torch.optim import AdamW
9  from transformers import AutoModelForCausalLM, AutoTokenizer,BitsAndBytesConfig
10
11 from xtuner.dataset import process_hf_dataset
12 from xtuner.dataset.collate_fns import default_collate_fn
13 from xtuner.dataset.map_fns import pretrain_map_fn
14 from xtuner.engine.hooks import (DatasetInfoHook, EvaluateChatHook,
15                                 VarlenAttnArgsToMessageHubHook)
16 from xtuner.engine.runner import TrainLoop
17 from xtuner.model import SupervisedFinetune
18
19 ##########################################################################
20 #                          PART 1  Settings                              #
21 ##########################################################################
22 # Model
23 pretrained_model_name_or_path = './internlm2_5-1_8b-chat'
24 use_varlen_attn = False
25
26 # Data
27 data_files = ['./datasets/pretrain.jsonl']
28 max_length = 2048
29 pack_to_max_length = False
30
31 # Scheduler & Optimizer
32 batch_size = 8  # per_device
33 accumulative_counts = 32  # bs = 1 GPU * 1 batch_size_per_device * 16 acc
34 dataloader_num_workers = 0
35 max_epochs = 1
36 optim_type = AdamW
37 lr = 2e-5
38 betas = (0.9, 0.999)
39 weight_decay = 0
40 max_norm = 1  # grad clip
41 warmup_ratio = 0.03
42
43 # Save
44 save_steps = 500
45 save_total_limit = 2  # Maximum checkpoints to keep (-1 means unlimited)
46
```

```python
# Evaluate the generation performance during the training
evaluation_freq = 500
SYSTEM = ''
evaluation_inputs = ['上海是', 'Shanghai is']

#######################################################################
#                      PART 2  Model & Tokenizer                      #
#######################################################################
tokenizer = dict(
    type=AutoTokenizer.from_pretrained,
    pretrained_model_name_or_path=pretrained_model_name_or_path,
    trust_remote_code=True,
    padding_side='right')

model = dict(
    type=SupervisedFinetune,
    use_varlen_attn=use_varlen_attn,
    llm=dict(
        type=AutoModelForCausalLM.from_pretrained,
        pretrained_model_name_or_path=pretrained_model_name_or_path,
        trust_remote_code=True,
        quantization_config=dict(
            type=BitsAndBytesConfig,
            load_in_4bit=True,
            load_in_8bit=False,
            llm_int8_threshold=6.0,
            llm_int8_has_fp16_weight=False,
            bnb_4bit_compute_dtype=torch.float16,
            bnb_4bit_use_double_quant=True,
            bnb_4bit_quant_type='nf4')
    ),
    lora=dict(
        type=LoraConfig,
        r=64,
        lora_alpha=16,
        lora_dropout=0.1,
        bias='none',
        task_type='CAUSAL_LM')
)


#######################################################################
#                      PART 3  Dataset & Dataloader                   #
#######################################################################
train_dataset = dict(
    type=process_hf_dataset,
    dataset=dict(type=load_dataset, path='json', data_files=data_files),
    tokenizer=tokenizer,
    max_length=max_length,
    dataset_map_fn=pretrain_map_fn,
    template_map_fn=None,
    remove_unused_columns=True,
    shuffle_before_pack=False,
    pack_to_max_length=pack_to_max_length,
    use_varlen_attn=use_varlen_attn)

train_dataloader = dict(
    batch_size=batch_size,
    num_workers=dataloader_num_workers,
    dataset=train_dataset,
    sampler=dict(type=DefaultSampler, shuffle=True),
    collate_fn=dict(type=default_collate_fn, use_varlen_attn=use_varlen_attn))
```

```python

#######################################################################
#                      PART 4  Scheduler & Optimizer                  #
#######################################################################
# optimizer
optim_wrapper = dict(
    type=AmpOptimWrapper,
    optimizer=dict(
        type=optim_type, lr=lr, betas=betas, weight_decay=weight_decay),
    clip_grad=dict(max_norm=max_norm, error_if_nonfinite=False),
    accumulative_counts=accumulative_counts,
    loss_scale='dynamic',
    dtype='float16')

# learning policy
# More information: https://github.com/open-mmlab/mmengine/blob/main/docs/en/tutorials/param_scheduler.md  # noqa: E501
param_scheduler = [
    dict(
        type=LinearLR,
        start_factor=1e-5,
        by_epoch=True,
        begin=0,
        end=warmup_ratio * max_epochs,
        convert_to_iter_based=True),
    dict(
        type=CosineAnnealingLR,
        eta_min=0.0,
        by_epoch=True,
        begin=warmup_ratio * max_epochs,
        end=max_epochs,
        convert_to_iter_based=True)
]

# train, val, test setting
train_cfg = dict(type=TrainLoop, max_epochs=max_epochs)

#######################################################################
#                            PART 5  Runtime                          #
#######################################################################
# Log the dialogue periodically during the training process, optional
custom_hooks = [
    dict(type=DatasetInfoHook, tokenizer=tokenizer),
    dict(
        type=EvaluateChatHook,
        tokenizer=tokenizer,
        every_n_iters=evaluation_freq,
        evaluation_inputs=evaluation_inputs,
        system=SYSTEM)
]

if use_varlen_attn:
    custom_hooks += [dict(type=VarlenAttnArgsToMessageHubHook)]

# configure default hooks
default_hooks = dict(
    # record the time of every iteration.
    timer=dict(type=IterTimerHook),
    # print log every 10 iterations.
    logger=dict(type=LoggerHook, log_metric_by_epoch=False, interval=10),
    # enable the parameter scheduler.
    param_scheduler=dict(type=ParamSchedulerHook),
```

```
    # save checkpoint per `save_steps`.
    checkpoint=dict(
        type=CheckpointHook,
        by_epoch=False,
        interval=save_steps,
        max_keep_ckpts=save_total_limit),
    # set sampler seed in distributed evrionment.
    sampler_seed=dict(type=DistSamplerSeedHook),
)

# configure environment
env_cfg = dict(
    # whether to enable cudnn benchmark
    cudnn_benchmark=False,
    # set multi process parameters
    mp_cfg=dict(mp_start_method='fork', opencv_num_threads=0),
    # set distributed parameters
    dist_cfg=dict(backend='nccl'),
)

# set visualizer
visualizer = None

# set log level
log_level = 'INFO'

# load from which checkpoint
load_from = None

# whether to resume training from the loaded checkpoint
resume = False

# Defaults to use random seed and disable `deterministic`
randomness = dict(seed=None, deterministic=False)

# set log processor
log_processor = dict(by_epoch=False)

```

## 6.2 模型转换pth_to_hf

Code block

```
xtuner convert pth_to_hf pretrain.py \
                         ./work_dirs/pretrain/iter_xx.pth \
                         ./work_dirs/pretrain/hf
```

## 6.3 模型合并merge

Code block

```
xtuner convert merge ./internlm2_5-1_8b-chat \
                     ./work_dirs/pretrain/hf \
                     ./work_dirs/pretrain/merged \
                     --max-shard-size 2GB

```

## 6.4 SFT训练

Code block

```
xtuner train sft.py --deepspeed deepspeed_zero1
```

注意替换模型和数据集的位置分别位于34行38行

Code block

```
pretrained_model_name_or_path = "./work_dirs/pretrain/merged"

alpaca_en_path = "./datasets/sftdata.jsonl"#可以换成自己的数据路径
```

sft.py

```
# Copyright (c) OpenMMLab. All rights reserved.
import torch
from datasets import load_dataset
from mmengine.dataset import DefaultSampler
from mmengine.hooks import (
    CheckpointHook,
    DistSamplerSeedHook,
    IterTimerHook,
    LoggerHook,
    ParamSchedulerHook,
)
from mmengine.optim import AmpOptimWrapper, CosineAnnealingLR, LinearLR
from peft import LoraConfig
from torch.optim import AdamW
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig

from xtuner.dataset import process_hf_dataset
from xtuner.dataset.collate_fns import default_collate_fn
from xtuner.dataset.map_fns import alpaca_map_fn, template_map_fn_factory
from xtuner.engine.hooks import (
    DatasetInfoHook,
    EvaluateChatHook,
    VarlenAttnArgsToMessageHubHook,
)
from xtuner.engine.runner import TrainLoop
from xtuner.model import SupervisedFinetune
from xtuner.parallel.sequence import SequenceParallelSampler
from xtuner.utils import PROMPT_TEMPLATE, SYSTEM_TEMPLATE

#######################################################################
#                          PART 1  Settings                           #
#######################################################################
# Model
pretrained_model_name_or_path = "./work_dirs/pretrain/merged"
use_varlen_attn = False

```

```python
# Data
alpaca_en_path = "./datasets/sftdata.jsonl"#换成自己的数据路径
prompt_template = PROMPT_TEMPLATE.internlm2_chat
max_length = 2048
pack_to_max_length = False

# parallel
sequence_parallel_size = 1

# Scheduler & Optimizer
batch_size = 1  # per_device
accumulative_counts = 1
accumulative_counts *= sequence_parallel_size
dataloader_num_workers = 0
max_epochs = 3
optim_type = AdamW
lr = 2e-4
betas = (0.9, 0.999)
weight_decay = 0
max_norm = 1  # grad clip
warmup_ratio = 0.03

# Save
save_steps = 500
save_total_limit = 2  # Maximum checkpoints to keep (-1 means unlimited)

# Evaluate the generation performance during the training
evaluation_freq = 500
SYSTEM = SYSTEM_TEMPLATE.alpaca
evaluation_inputs = ["请给我介绍五个上海的景点", "Please tell me five scenic spots in Shanghai"]

#######################################################################
#                      PART 2  Model & Tokenizer                      #
#######################################################################
tokenizer = dict(
    type=AutoTokenizer.from_pretrained,
    pretrained_model_name_or_path=pretrained_model_name_or_path,
    trust_remote_code=True,
    padding_side="right",
)

model = dict(
    type=SupervisedFinetune,
    use_varlen_attn=use_varlen_attn,
    llm=dict(
        type=AutoModelForCausalLM.from_pretrained,
        pretrained_model_name_or_path=pretrained_model_name_or_path,
        trust_remote_code=True,
        torch_dtype=torch.float16,
        quantization_config=dict(
            type=BitsAndBytesConfig,
            load_in_4bit=True,
            load_in_8bit=False,
            llm_int8_threshold=6.0,
            llm_int8_has_fp16_weight=False,
            bnb_4bit_compute_dtype=torch.float16,
            bnb_4bit_use_double_quant=True,
            bnb_4bit_quant_type="nf4",
        ),
    ),
    lora=dict(
        type=LoraConfig,
```

```python
        r=64,
        lora_alpha=16,
        lora_dropout=0.1,
        bias="none",
        task_type="CAUSAL_LM",
    ),
)

#######################################################################
#                      PART 3  Dataset & Dataloader                   #
#######################################################################
alpaca_en = dict(
    type=process_hf_dataset,
    dataset=dict(type=load_dataset, path='json', data_files=alpaca_en_path),
    tokenizer=tokenizer,
    max_length=max_length,
    dataset_map_fn=alpaca_map_fn,
    template_map_fn=dict(type=template_map_fn_factory, template=prompt_template),
    remove_unused_columns=True,
    shuffle_before_pack=True,
    pack_to_max_length=pack_to_max_length,
    use_varlen_attn=use_varlen_attn,
)

sampler = SequenceParallelSampler if sequence_parallel_size > 1 else DefaultSampler
train_dataloader = dict(
    batch_size=batch_size,
    num_workers=dataloader_num_workers,
    dataset=alpaca_en,
    sampler=dict(type=sampler, shuffle=True),
    collate_fn=dict(type=default_collate_fn, use_varlen_attn=use_varlen_attn),
)

#######################################################################
#                     PART 4  Scheduler & Optimizer                   #
#######################################################################
# optimizer
optim_wrapper = dict(
    type=AmpOptimWrapper,
    optimizer=dict(type=optim_type, lr=lr, betas=betas, weight_decay=weight_decay),
    clip_grad=dict(max_norm=max_norm, error_if_nonfinite=False),
    accumulative_counts=accumulative_counts,
    loss_scale="dynamic",
    dtype="float16",
)

# learning policy
# More information: https://github.com/open-mmlab/mmengine/blob/main/docs/en/tutorials/param_scheduler.md  # noqa: E501
param_scheduler = [
    dict(
        type=LinearLR,
        start_factor=1e-5,
        by_epoch=True,
        begin=0,
        end=warmup_ratio * max_epochs,
        convert_to_iter_based=True,
    ),
    dict(
        type=CosineAnnealingLR,
        eta_min=0.0,
        by_epoch=True,
```

```
        begin=warmup_ratio * max_epochs,
        end=max_epochs,
        convert_to_iter_based=True,
    ),
]

# train, val, test setting
train_cfg = dict(type=TrainLoop, max_epochs=max_epochs)

#######################################################################
#                          PART 5  Runtime                            #
#######################################################################
# Log the dialogue periodically during the training process, optional
custom_hooks = [
    dict(type=DatasetInfoHook, tokenizer=tokenizer),
    dict(
        type=EvaluateChatHook,
        tokenizer=tokenizer,
        every_n_iters=evaluation_freq,
        evaluation_inputs=evaluation_inputs,
        system=SYSTEM,
        prompt_template=prompt_template,
    ),
]

if use_varlen_attn:
    custom_hooks += [dict(type=VarlenAttnArgsToMessageHubHook)]

# configure default hooks
default_hooks = dict(
    # record the time of every iteration.
    timer=dict(type=IterTimerHook),
    # print log every 10 iterations.
    logger=dict(type=LoggerHook, log_metric_by_epoch=False, interval=10),
    # enable the parameter scheduler.
    param_scheduler=dict(type=ParamSchedulerHook),
    # save checkpoint per `save_steps`.
    checkpoint=dict(
        type=CheckpointHook,
        by_epoch=False,
        interval=save_steps,
        max_keep_ckpts=save_total_limit,
    ),
    # set sampler seed in distributed evrionment.
    sampler_seed=dict(type=DistSamplerSeedHook),
)

# configure environment
env_cfg = dict(
    # whether to enable cudnn benchmark
    cudnn_benchmark=False,
    # set multi process parameters
    mp_cfg=dict(mp_start_method="fork", opencv_num_threads=0),
    # set distributed parameters
    dist_cfg=dict(backend="nccl"),
)

# set visualizer
visualizer = None

# set log level
log_level = "INFO"
```

```

# load from which checkpoint
load_from = None

# whether to resume training from the loaded checkpoint
resume = False

# Defaults to use random seed and disable `deterministic`
randomness = dict(seed=None, deterministic=False)

# set log processor
log_processor = dict(by_epoch=False)

```

## 6.5 模型转换pth_to_hf

Code block

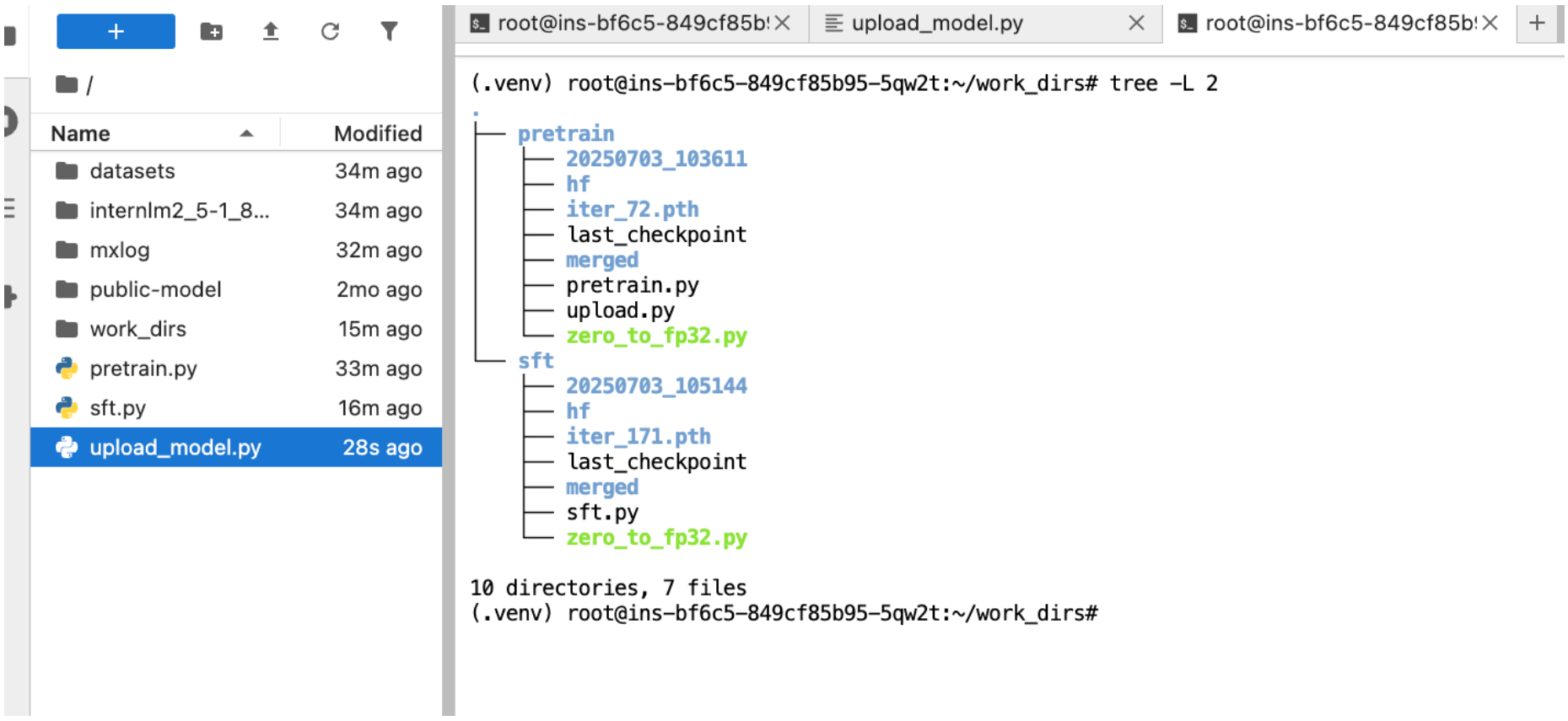
```
xtuner convert pth_to_hf sft.py \
                         ./work_dirs/sft/iter_xx.pth \
                         ./work_dirs/sft/hf
```

## 6.6 模型合并

Code block

```
#与原始模型合并
xtuner convert merge ./internlm2_5-1_8b-chat \
                     ./work_dirs/sft/hf \
                     ./work_dirs/sft/merged \
                     --max-shard-size 2GB
```

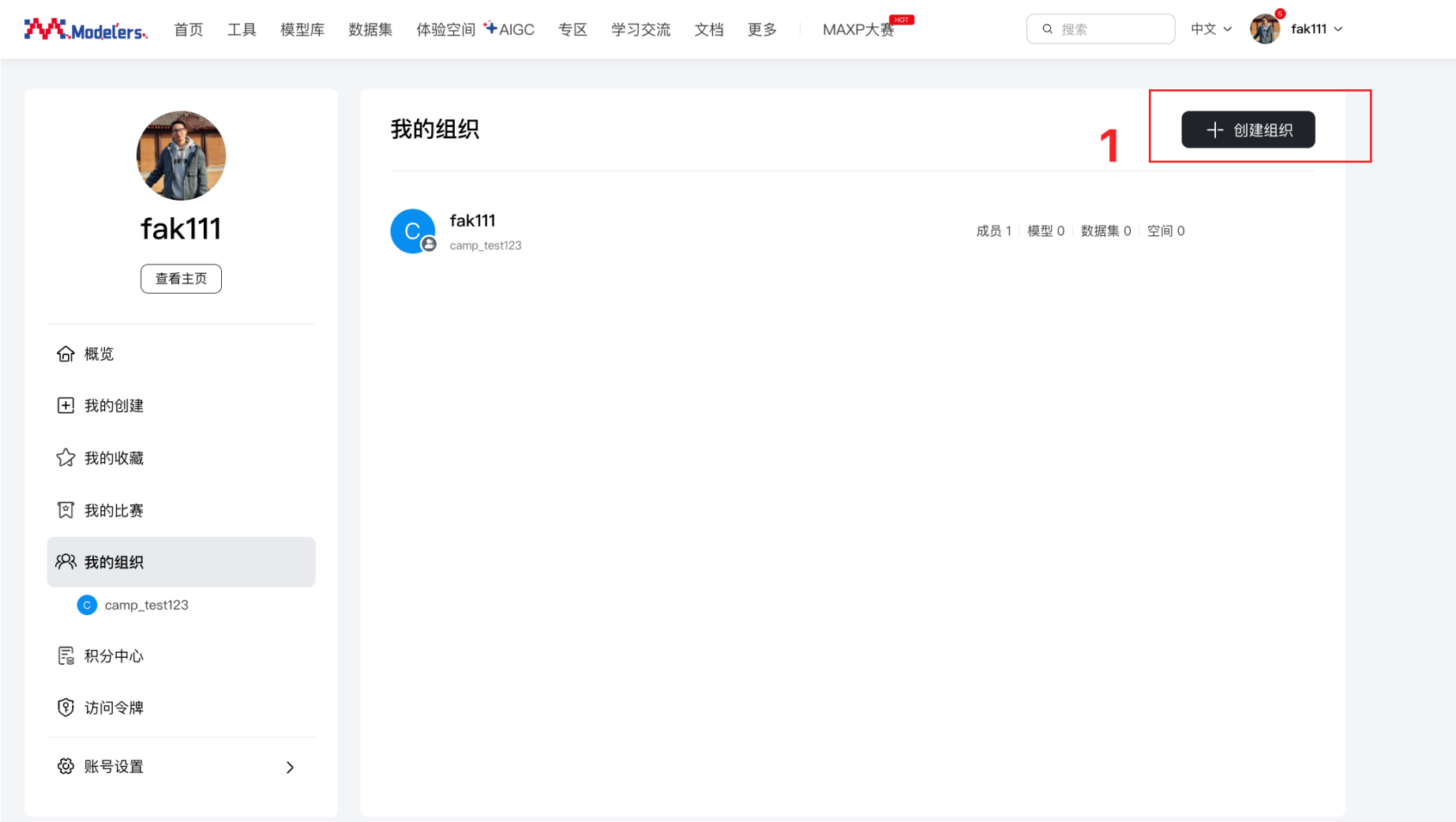
训练完后 `work_dirs` 目录结构参考：



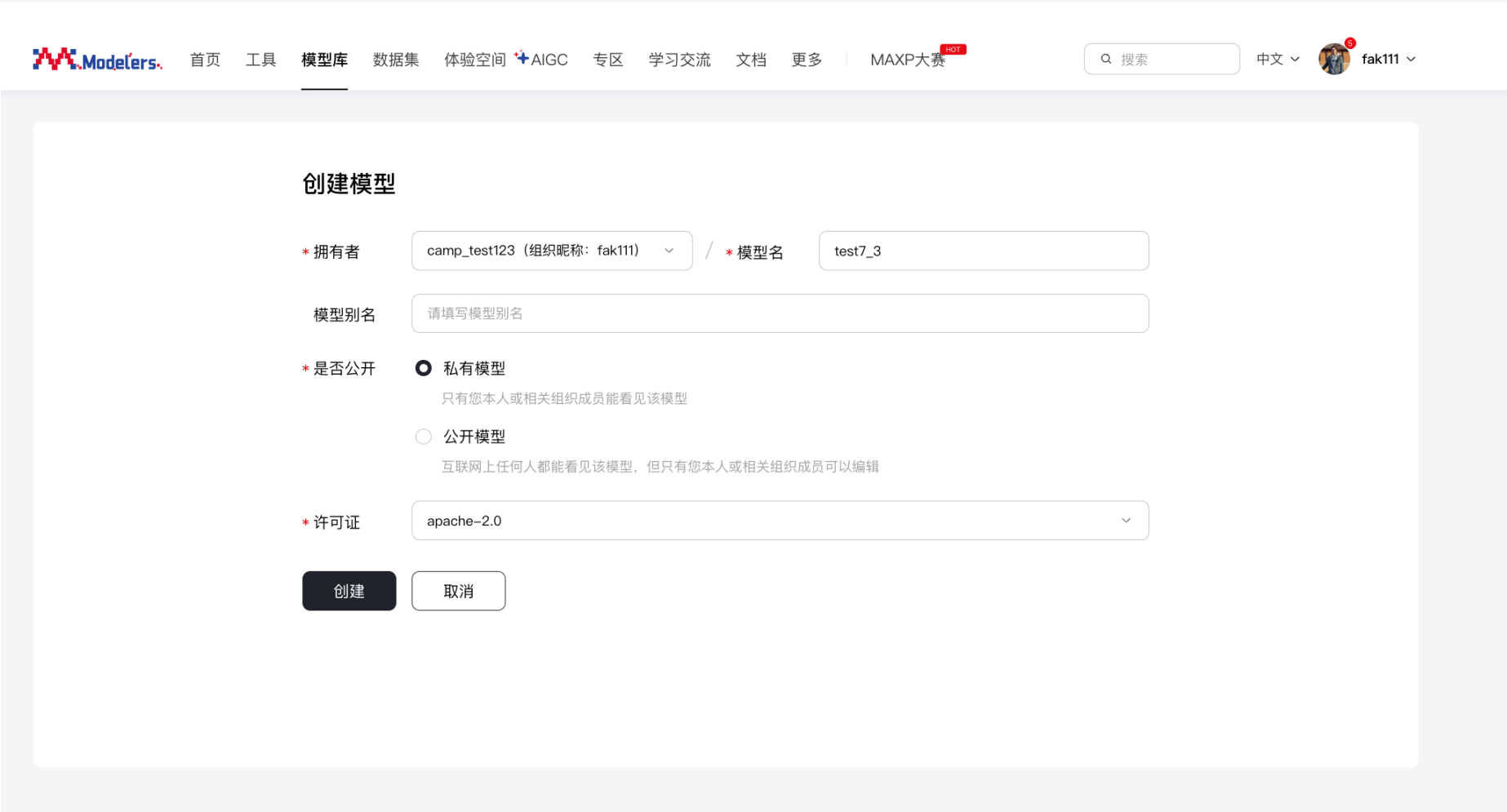

# 7. 上传到魔乐

## 7.1 魔乐上传配置

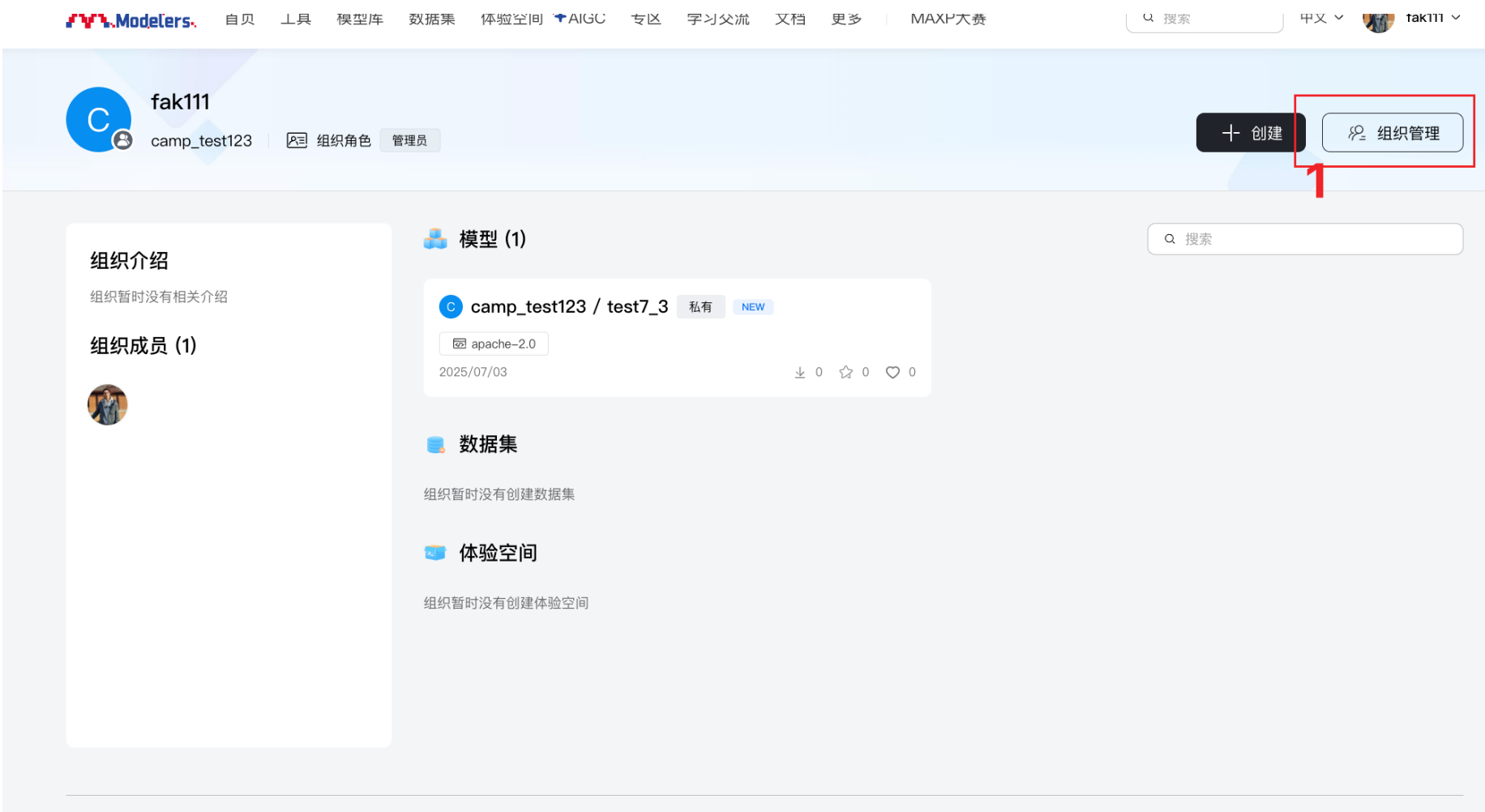
1. 创建组织



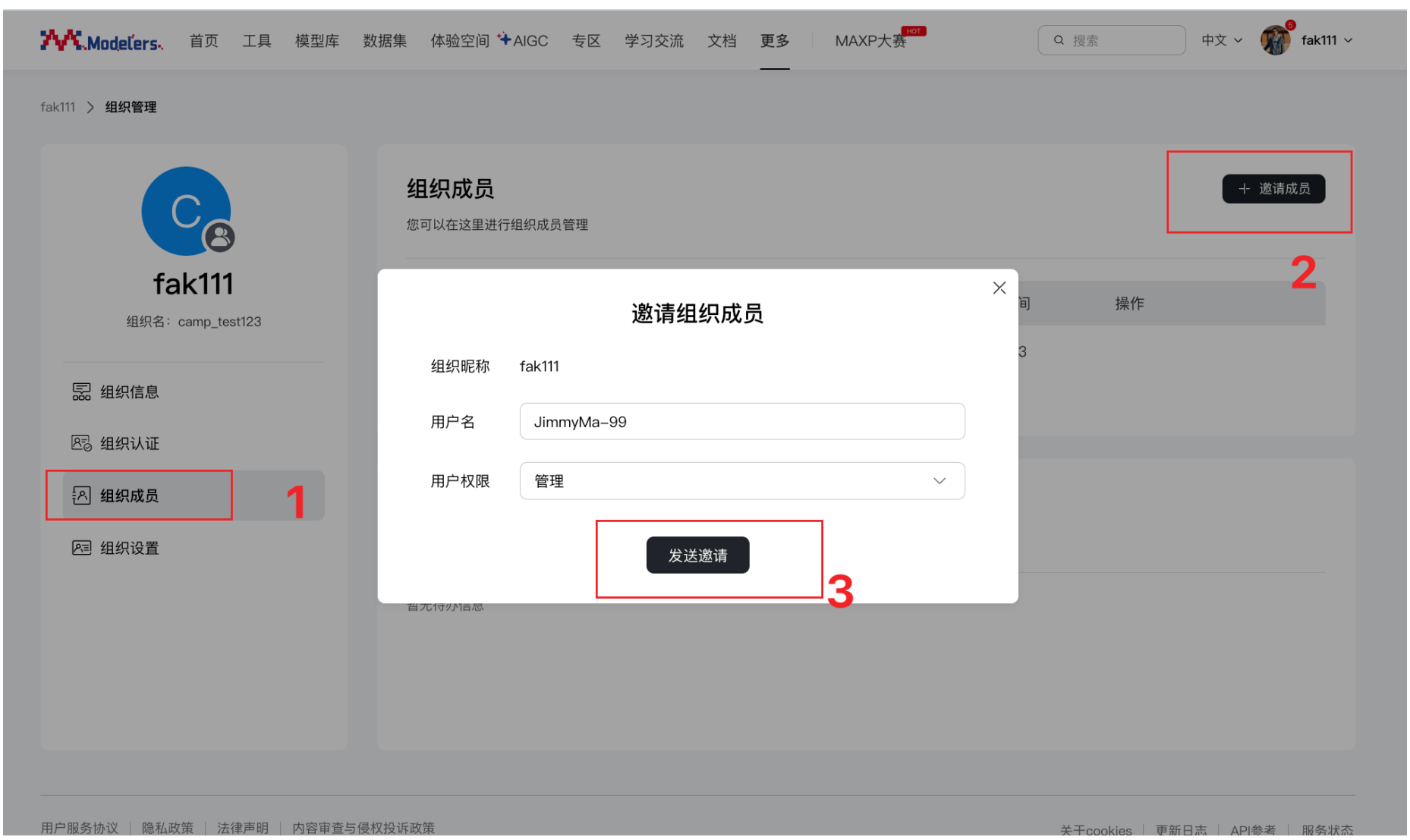

2. 在组织内创建模型




3. 为了保证比赛公正，本次使用的魔乐社区托管权重，请使用 `组织` 功能，并邀请 `JimmyMa-99` 用户加入团队，同时请将上传的是否公开为 `私有模型` 。




4. 一定要邀请米佬，等同意 。如在 24 小时内没有同意邀请，请联系微信`小镇数字青年`




## 7.2 上传模型

1. 下载所需环境包

pip 下载环境包

```
python3.10 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install openmind_hub[all]
```

```
pip install "openmind[pt]" --extra-index-url https://download.pytorch.org/whl/cpu
```

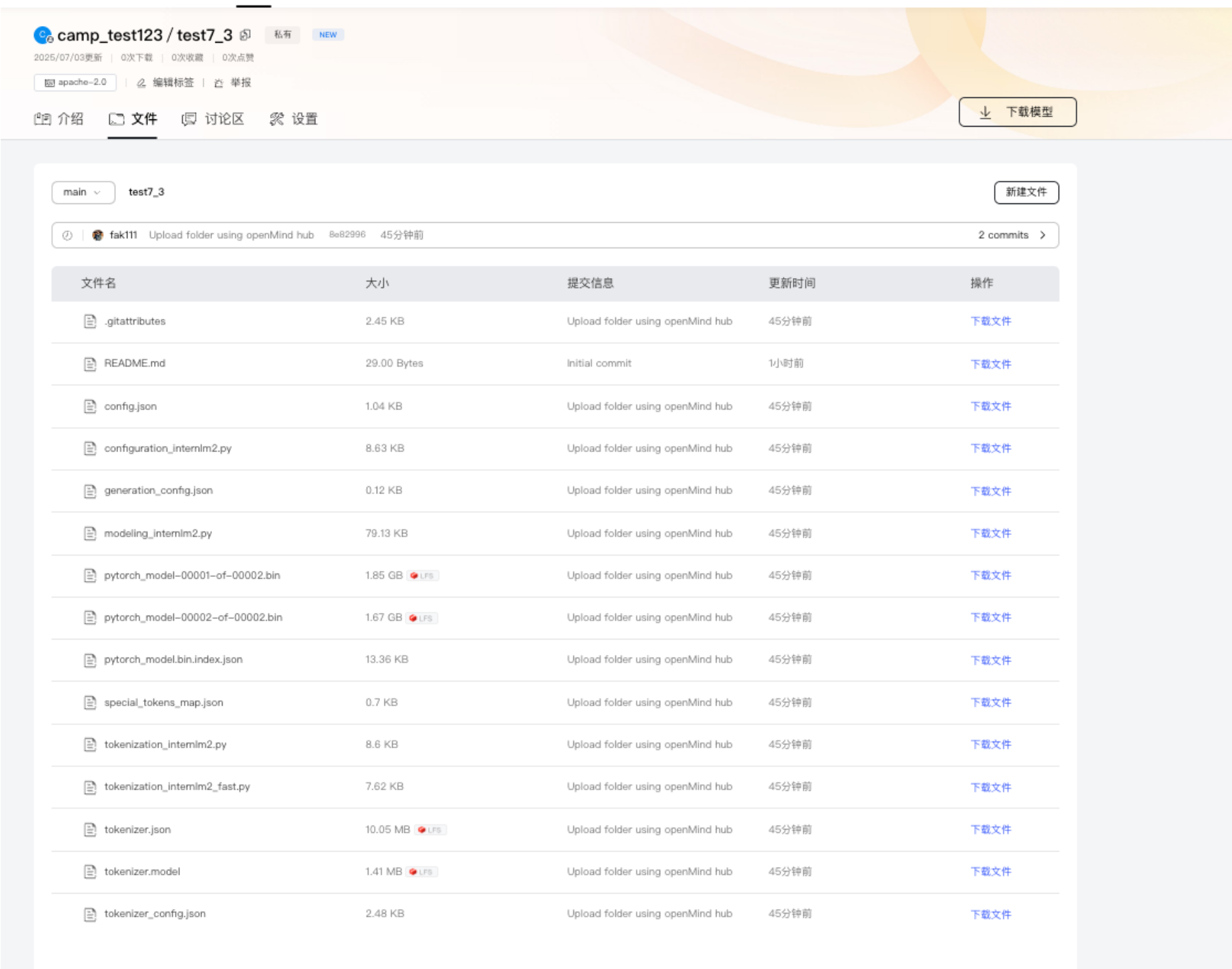
2. 运行上传脚本

upload_model.py

```
from openmind_hub import upload_folder

upload_folder(
    token="a5cxxe",
    folder_path="/root/work_dirs/sft/merged",#模型位置
    repo_id="xxx",#模型空间名字
)
```

魔乐空间会如如下所示：



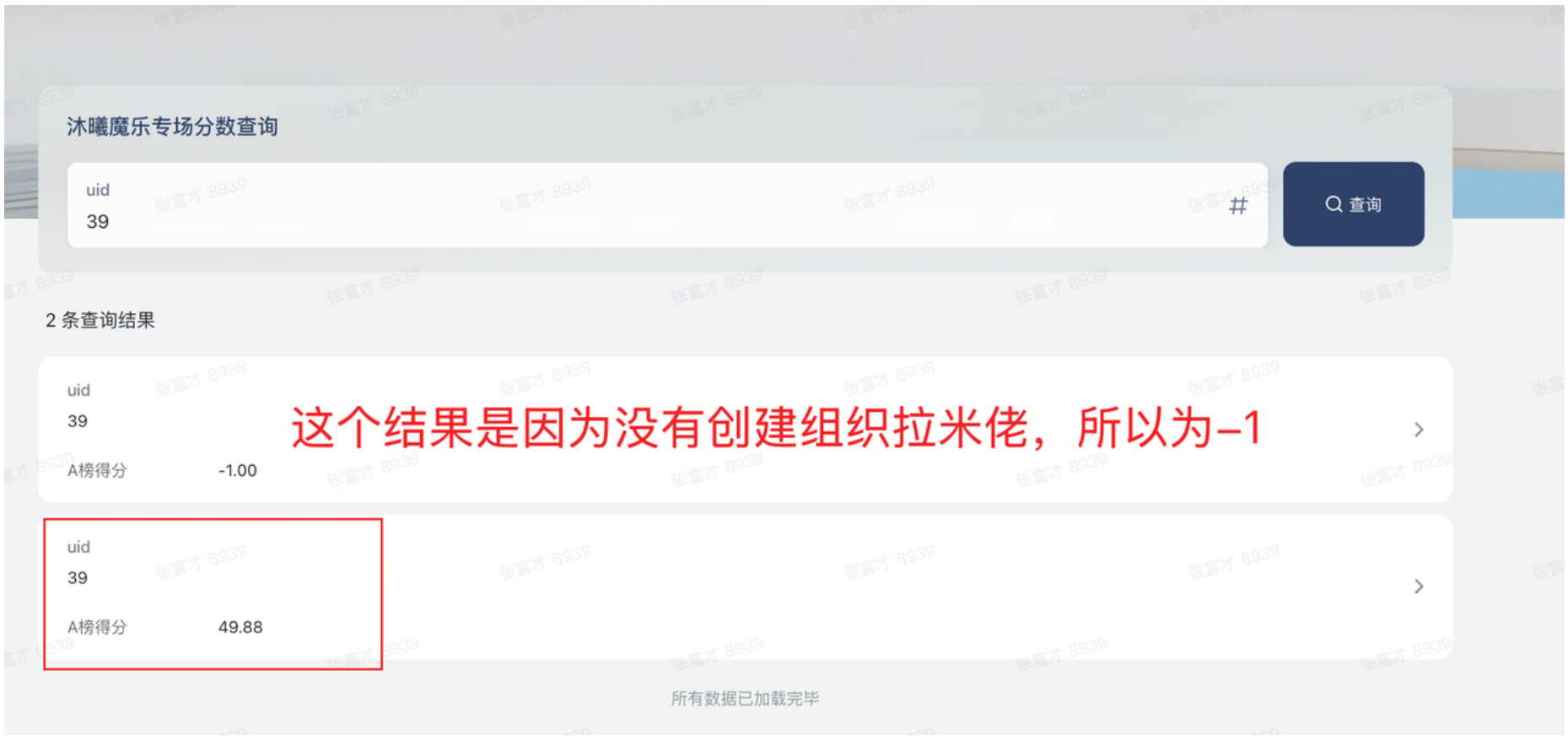

## 7.3 提交评测

1. 确保 `JimmyMa-99` 用户加入团队

2. 提交测评的入口：https://aicarrier.feishu.cn/share/base/form/shrcnCR6hPPtl3v3y2ZVXomhME9，输入 uid 和对应的微调权重的模型 ID。

提交完成，自动等评测榜单更新(大概等十分钟)。

查询测评的分数入口：https://aicarrier.feishu.cn/share/base/query/shrcn3UPdvR9e0D1REcjQ9LLNub 输入 uid 即可查询。




## 8. 本文参考

L1G4-InternLM 论文分类实践（沐曦版）